

Optimization of Lightweight Polymer Composites Using Finite Element Analysis Machine Learning and Topology Optimization Techniques for Aerospace Applications

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Abstract

The advancement of aerospace engineering depends on lightweight polymer matrix composites (PMCs) because they help decrease weight while improving fuel efficiency and payload capacity together with increased structural integrity. Research developed a computer program comprising FEA with ANN and TO optimize high-performance PMCs through integrated design approaches. The combination of Python-controlled LS-DYNA simulations measured hybrid composite laminate resistance to impact while an ANN model obtained data from simulations to forecast projectile velocities across different reinforcement patterns. The utilization of TO allowed researchers to reduce material mass by 28% while maintaining 92% of stiffness properties and increasing energy absorption by 25%. The research established that Kevlar/Epoxy and hybrid composite materials outperform other options as aerospace blocking components due to their excellent ability to resist impacts. The study proves that using computational intelligence and optimization techniques will produce effective aerospace-grade composites in the next generation.

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INTRODUCTION

The aerospace industry speeds up its development through continuous advancement of performance and fuel consumption improvement and environmental protection strategies. The industry depends on lightweight materials for these goals because weight losses result in improved cargo capacity which reduces fuel consumption and raises operational effectiveness [1]. Livermore Laboratory reports that PMCs represent a revolutionary choice because they provide specific engineering requirements such as exceptional strength-to-weight ratios and thermal stability along with property adjustments [2]. Made of a sequential polymer matrix containing particles or fibers that provide mechanical stability and light-weight properties for use in the aerospace industry in fuselage and turbine blade sections and suspension elements [3]. PMCs deliver more than their weight-saving abilities